FORMULA

CHAPTER I : Concepts

$$\text{Precision} = \frac{TP}{TP+FP} \quad ; \quad \text{Accuracy} = \frac{TP+TN}{N} \quad ; \quad \text{Recall} = \frac{TP}{TP+FN} = \text{Sensitivity} = TPR$$

$$\text{Specificity} = \frac{TN}{TN+FP} = TNR \quad ; \quad FPR = \frac{FP}{FP+TN} = 1 - \text{Specificity}$$

$F1 = 2\frac{Pre \cdot Rec}{Pre + Rec}$ | Pre cao: ít FP ; Rec cao: ít FN ; F1: balance P/R ; Macro-F1: avg of F1 of classes
Weighted-F1: weighted F1 over classes ; Acc mislead for imbalance data

- ROC-curve: x = FPR, y = TPR
- PR-curve: x = Rec, y = Pre
- ROC-AUC: rank positive > negative

Underfit (High bias) → Incr model complexity, better feature, longer train
Overfit (High variance) → Regularization, pruning, early stop, more data
→ Goal: generalization

- Rare positives → PR-curve show clear FP problem and precision than ROC

CHAPTER II: Decision Trees

Entropy: $H(S) = -\sum_{k} p_k \cdot \log_2 p_k$ $\quad 0 < H(S) < 1$

Info Gain: $IG(S,A) = H(S) - \sum_{v} \frac{|S_v|}{S} H(S_v)$: pick split with larger IG

Gini: $\text{Gini}(S) = 1 - \sum_{k} p_k^2$ $\quad 0 < \text{Gini}(S) < 1$

Weighted Gini: $\text{Gini}_{split} = \sum_{v} \frac{|S_v|}{|S|} \text{Gini}(S_v)$: pick split with smaller Gini

Regression Tree:
- Dự đoán ở lá thường là avg các target trong lá
- Split: ↓ SSE or variance

- Classification Tree: Predict class.
- Regression tree: Predict value

Split Info: $\text{SplitInfo}(S,A) = -\sum \frac{|S_v|}{|S|} \log_2 \frac{|S_v|}{|S|}$

Gain Ratio: $\text{GainRatio}(S,A) = \frac{IG(S,A)}{\text{SplitInfo}(S,A)}$ → Reduce bias of IG for attr with many values. If SplitInfo = 0, the split provides no useful partition and the ratio is undefined

ID3 - IG - Preprocess missing values.
C4.5 - GR - Fractional examples
CART - Gini - Surrogated splits

- Preprune: Prune when a condition is met
- Postprune: Remove branches w/o much improvement
- Cost-complexity prune: $R_\alpha(T) = R(T) + \alpha|T|$

$R(T)$: độ lỗi / không hợp của cây
$|T|$: complexity, often num of leaves.
$\alpha \uparrow \rightarrow$ ↑penalty for complex tree

CHAPTER III: Neural Networks :v

A.

Linear Regression: $\hat{y} = w^T \cdot x + b$. For full data, can merge bias into w^T by adding $x_0 = 1$
Squared-error loss: $E(w) = \frac{1}{2}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2$. If use mean squared error (MSE): $\times \frac{1}{N}$

Normal Equation: $w = (X^T X)^{-1} X^T y$

Gradient Descent: $w \leftarrow w - \eta \nabla_w E(w)$

For linear reg: $\frac{\partial E}{\partial w} = X^T(Xw - y)$ if $E = \frac{1}{2}\|Xw - y\|^2$. If loss is mean → $\times \frac{1}{N}$

Feature scaling: Standardization: $x' = \frac{x - \mu}{s}$, Min-max scaling: $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$

↳ Calc on train → transform val/test → Help GD faster and more stable

Logistic Regression: $z = w^T x + b \rightarrow \hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$; $\sigma'(z) = \sigma(z)\cdot(1 - \sigma(z))$
Binary Cross-Entropy: $L = -y \cdot \ln\hat{y} - (1-y)\ln(1-\hat{y})$
Gradient: $\frac{\partial L}{\partial w_i} = (\hat{y} - y)\, x_i$; $\frac{\partial L}{\partial b} = \hat{y} - y$ $\rightarrow$ $w_i \leftarrow w_i - \eta(\hat{y} - y)\cdot x_i$

L2 Regularization: $L_{reg} = L + \frac{\lambda}{2}\|w\|^2$. Gradient add λw if not regu bias. $\nabla L_{reg} = \nabla L + \lambda w$
↳ Penalize large weights → Overfit. L1 more sparse solutions, L2 near 0

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B.

Perceptron: $z = w^T x + b$ → Use activation to classify. $y, \hat{y} \in \{0; 1\}$.

Update rule: $w_i \leftarrow w_i + \eta(y - \hat{y})x_i$, $b \leftarrow b + \eta(y - \hat{y})$

+ Correct → no update
+ Pre: 0, Act: 1 → adjust weight to lessen if feature (+)
+ Pre: 1, Act: 0 → ____ lessen if feature (-)

} → Converges after finite step if data linearly separable and condition of perceptron convergence theorem met

Delta rule: Use linear unit, with continuous output: $o = w^T x$.

Squared error: $E = \frac{1}{2}(t - o)^2$

Gradient: $\partial E / \partial w_i = -(t - o) x_i$

Update: $w_i \leftarrow w_i + \eta(t - o)x_i$, $b \leftarrow b + \eta(t - o)$

Perception rule	Delta rule	
Nguồng	Linear, continuous	output
Phân loại sai	$t - o$	Sai số
Khi classify sai	Khi sai số $\neq 0$	update
Quy tắc sửa lỗi	Gradient sgn error	cập số

Stochastic Delta rule: $\Delta w_i = \eta(t - o)x_i$

Batch: $\Delta w_i = \eta \sum_d (t_d - o_d) x_{di}$

MLP:

Forward pass ~~của~~ lớp l:

$z^{(l)} = W^{(l)} h^{(l-1)} + b^{(l)}$

$h^{(l)} = \phi(z^{(l)})$, $h^{(0)} = x$

W: weight matrix
b: vector bias
z: giá trị trước activ
h: out layer trước
ϕ: activ func

Sigmoid: $\sigma(z) = 1/(1+e^{-z}) \rightarrow \sigma'(z) = \sigma(z)(1-\sigma(z))$ ✓

$\tanh(z) \rightarrow 1 - \tanh^2(z)$ ✗

ReLU: $\max(0, z) \rightarrow 0$ if $z < 0$, 1 if $z > 0$ ✓

Leaky ReLU: z if $z \geq 0$, αz if $z < 0 \rightarrow 1$ or α ✗

Backpropagation:

Forward tính: $z, h, \hat{y}$
→ Tính Loss L
→ Tính δ ở output
→ Truyền δ về hidden
→ Update w và b

$\delta_k = o_k(1 - o_k)(t_k - o_k)$ for output sgmoid with squared-error loss.

For hidden neuron h: $\delta_h = o_h(1 - o_h) \sum_k w_{kh} \delta_k$

Update: $\Delta w_{ji} = \eta \delta_j x_i$; $w_{ji} \leftarrow w_{ji} + \Delta w_{ji}$

bias update same way by $x_0 = 1 \rightarrow \Delta b_j = \eta \delta_j$

Momentum:

$\Delta w_{ji}^{(n)} = \eta \delta x + \alpha \Delta w^{(n-1)}$

$w^{(n)} = w^{(n-1)} + \Delta w^{(n)}$

α: momentum coeff.

init $\Delta w^{(0)} = 0$.

$\sigma(z) = \frac{1}{1+e^{-z}} \rightarrow \frac{d}{dz}[u(z)]^{-1} = -[u(z)]^{-2} \cdot u'(z)$

$\frac{d}{dz}(1 + e^{-z}) = -e^{-z} \rightarrow \sigma'(z) = -(1+e^{-z})^{-2}(-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2}$

$\left.\begin{aligned} \sigma(z) &= \frac{1}{1+e^{-z}} \\ 1 - \sigma(z) &= \frac{e^{-z}}{1+e^{-z}} \end{aligned}\right\} \rightarrow \sigma(z)[1 - \sigma(z)] = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \frac{e^{-z}}{(1+e^{-z})^2}$ → $\boxed{\sigma'(z) = \sigma(z)(1 - \sigma(z))}$

$w_j \rightarrow z \rightarrow \hat{y} \rightarrow L \Rightarrow \frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w_j}$

$L = -y \ln \hat{y} - (1-y)\ln(1-\hat{y}) \rightarrow \frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = \frac{-y(1-\hat{y}) + (1-y)\hat{y}}{\hat{y}(1-\hat{y})} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$

$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-y)$

$\frac{\partial z}{\partial w_j} = x_j$

} → $\frac{\partial L}{\partial w_j} = (\hat{y} - y)x_j$

$\frac{\partial z}{\partial b} = 1 \Rightarrow \frac{\partial L}{\partial b} = \hat{y} - y$

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